



MedGov AI MCP



Problem / Context

Healthcare systems face structural challenges that motivate these kind of works, so that we can make these difficulties more approachable.

Integration complexity

M x N problem: M hospital applications x N services = M x N integrations
(Leads to harder migrations and communication issues)

Logs and context

In medical context we need to know what was done, why it was done.

Analysis and patients

There's a lot of patients and a lot of documents going around, as well as analysis results that can be amplified with the use of AI for sometimes more accurate and consistent results.

Objective / Solution

After exploring the technologies of MCP and Agentic IA, as well as keeping up with current, in development technologies, we were guided towards 3 major use cases (with adjustments made through out the process and new ideas developed)



01

Use case 1 (Main focus)

A radiologist processes multiple daily exams. MONAI segments organs, RadLex provides report templates, the agent assembles a radiology report (possibly augmented with patient history)

02

Use case 2

Clinical notes system, with integration from other systems, for example getting medical consultation or picking and organizing slots.

03

Use case 3

Investigation, research platform

As said, adjustments were made along the way, and we tried to generalize the implementations to allow more use cases and scaling



Key concepts /Project

- 01

MCP
Lets LLM call external tools (MONAI, FHIR, RADLEX] through an uniform interface
- 02

Agentic AI
LLM (orchestrator) decides which tools to call, in which order, based on context and description of tools (No hardcoded workflow - different from what we had before)
- 03

DICOM
We decided to focus on this standard format for medical images, due to the extraction of metadata in order to run the inferences with the MONAI models. (There were a lot of problems trying to get data from other types of files, but we also accept .nii.gz)
- 04

RadLex / RadReport
Standardized radiology terminology and report templates
- 05

FHIR
HL7's standard for electronic health records, let's the agent pull patient demographics (we didn't focus too much on the patient part for now, but we did integrate sessions and patient context)
- 06

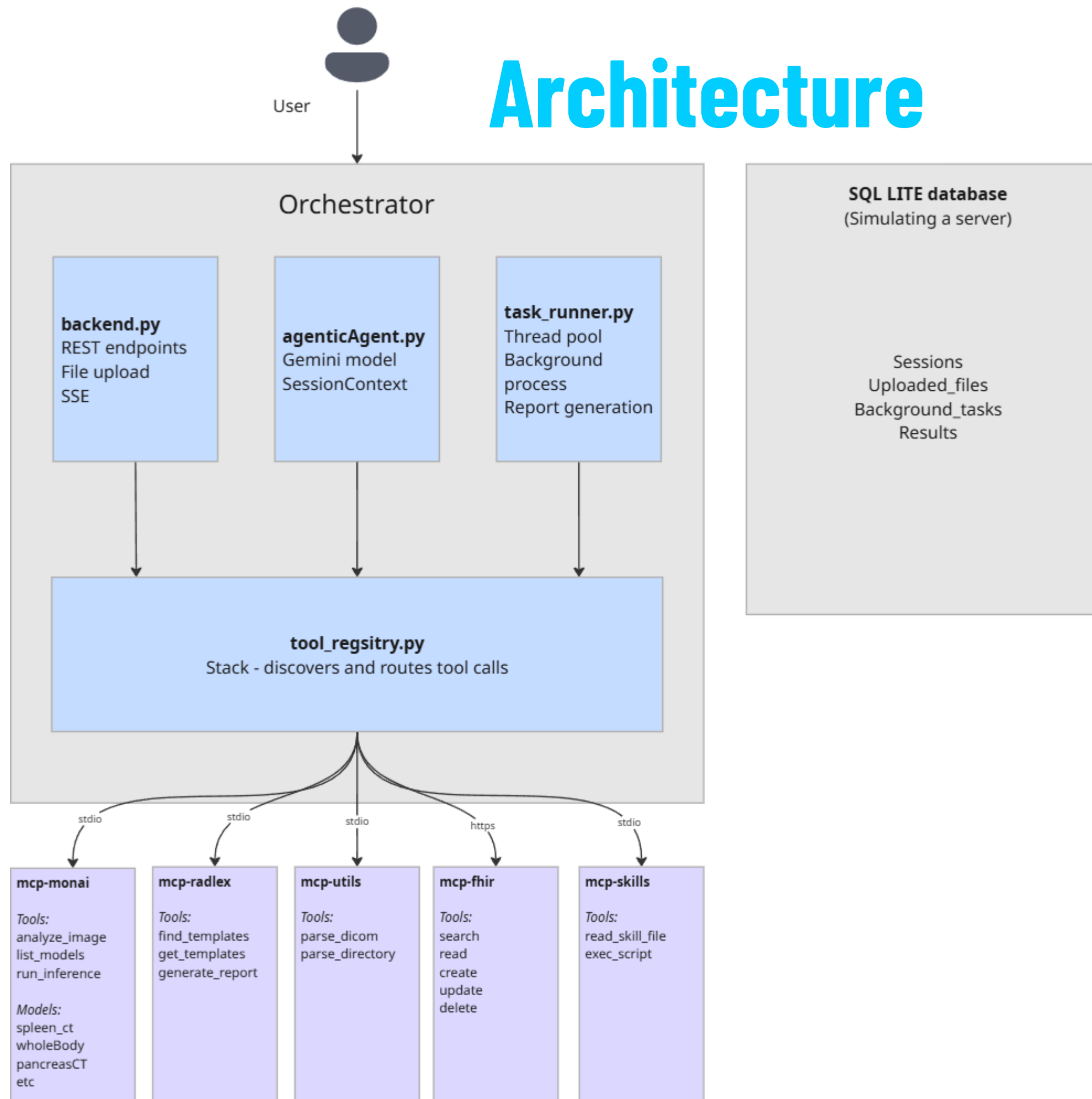
MONAI
Medical AI framework, host to pre-trained models
- 07

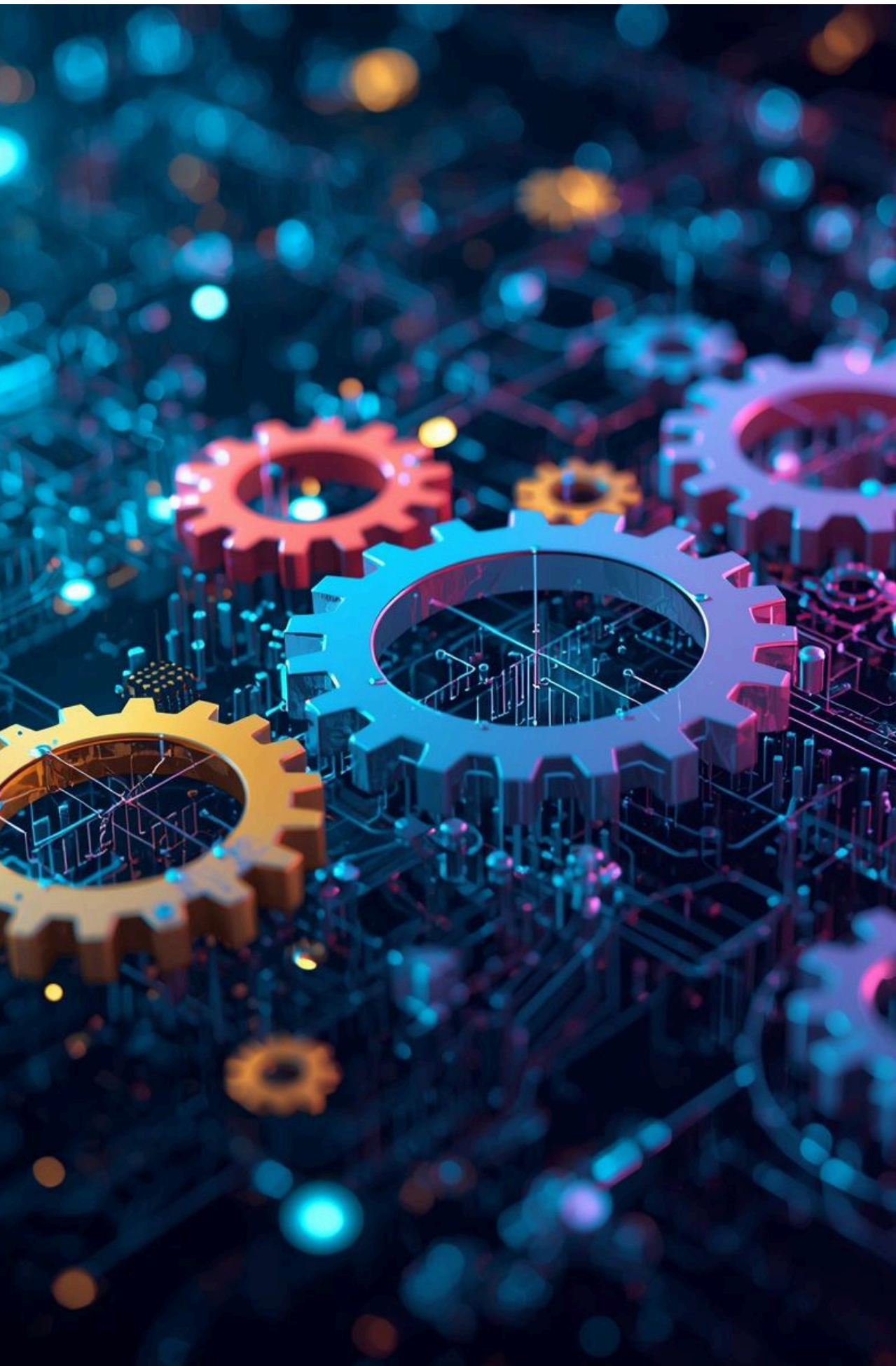
Background task queue
Generalized for anything that can be considered a background task in order to run it in the background while keeping the orchestrator active
- 08

SSE (Server sent events)
Real time push from backend to frontend (for notifications and on the run integration)
- 09

Session context
Self built for local models, but there's a toggle in the API itself from Gemini that were using

Architecture





Skills system

01

Motivation

First designed as a way to counter the bad choices that the local Ollama model was making, by specifying behaviors for the model for each tool that we wanted.

03

Scripts

It allows also for the execution of scripts within that skills

05

Where to use skills

- When there are no MCP tools for the task.
- When the context window scaling is needed (when there's too many MCP tools reading all descriptions and choosing appropriate one, takes longer than choosing a behavior type for the agent and then executing a skill within that behavior)

02

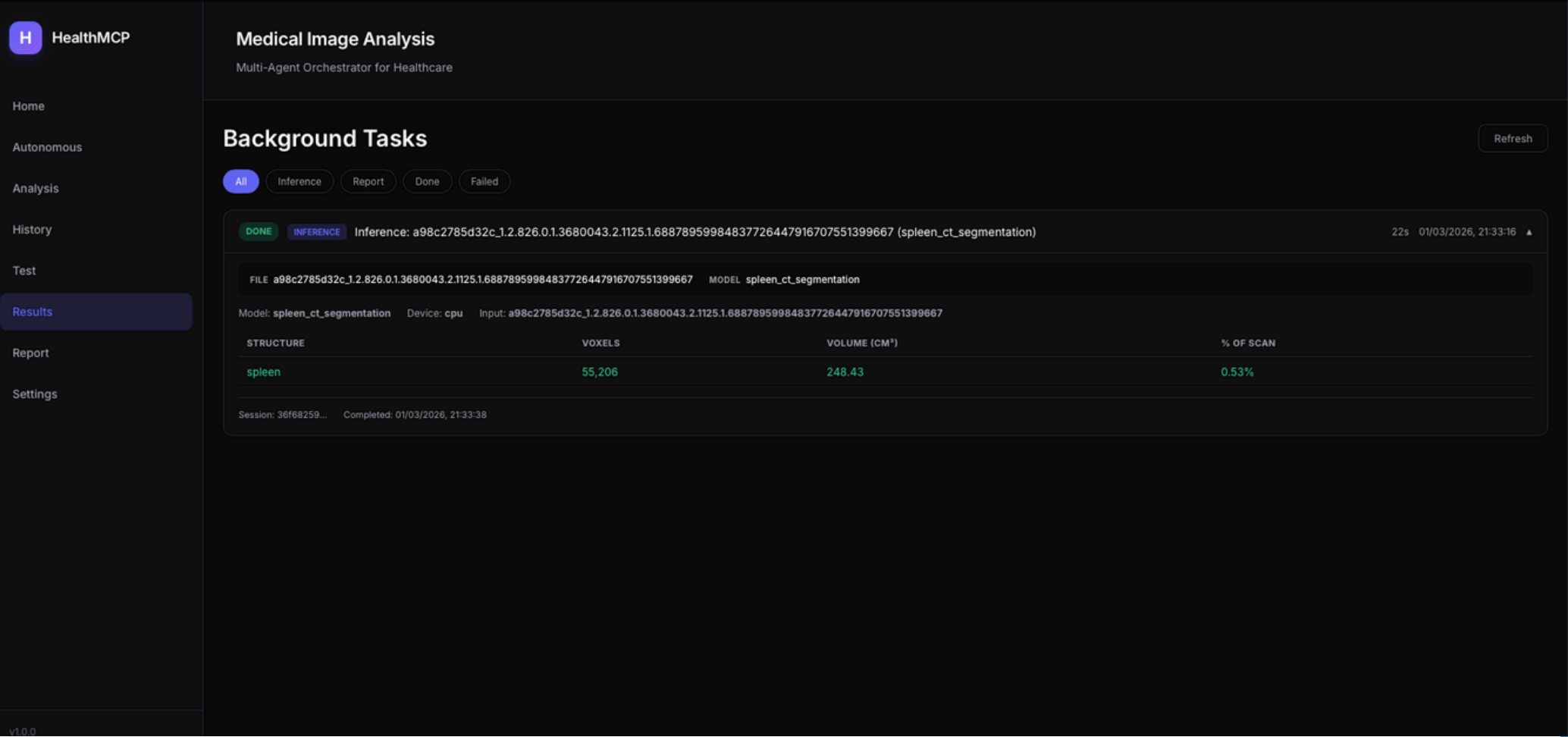
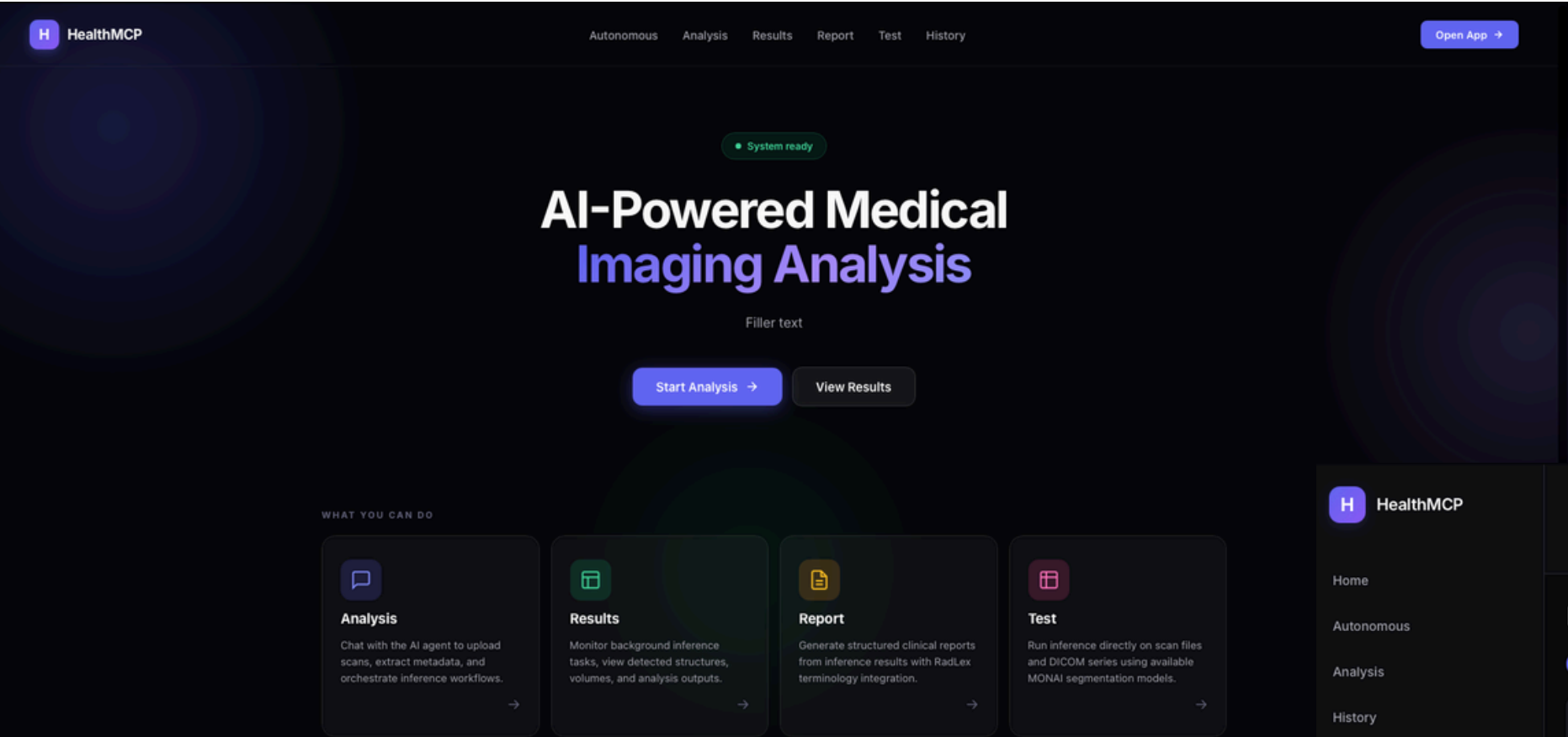
How it works

The model sees the directory of skill names, plus short descriptions, at the same time it loads .md files that have basically behavioral prompts.

04

Limitations

Skills add more tool calls per task, where a direct MCP tool does it in 1.
To extract DICOM data:
read_skill_file → *read references* → *execute script*
Where with MCP it reads descriptions for tools and executes *utils.parse_dicom*



FRONTEND

We took more of an application approach due to it being easier to understand specific goals, we also want your opinion on it.

Future work

Directory analyzer

A passive directory analyzer that detects background changes and keeps running inferences on the documents in that directory, according to what the orchestrator finds best to do with them.

Multi agent architecture

Possibility

Autonomous Agent

Instead of talking with the agent, have the agent work autonomously

Drag and drop

Just a drag and drop syntax that uploads all files with no chat bot and we just receive the results of whatever files were analyzed

Downloadable reports

And other files, be able to export them

Better error handling

Right now we detect errors on inferences and try to return what's wrong in a readable way to the user

Data anonymization agent

As part of the workflow for verification (Was done previously as POC so it can just be integrated)

CHALLENGES

Main challenges right now are mostly these 3:

Dealing with different types of data

Trying to expand this beyond .dcm files for inferences, and find other use cases in order to setup the acceptance or use of other types of file like pdf reports for example...

Optimization

Optimization of queries and code, for example: when running an inference on several files the model also runs several times the same tool call for analyze images for each file or directory passed as input.
Find a way to reduce the number of tokens spent per prompt.
Optimize logging and error handling for the same outcome

MONAI's segmentation models expectations

Finding ways to implement other types of models besides the MONAI ones, while also finding ways to make the already implemented ones work with different types of formats for other scans realized.

CHALLENGES

Writing report using MCP RadLex/Radreport

Other renal findings	
Adrenals	
Direct invasions by renal mass	
☐ no	
☐ indeterminate	
☐ yes	
Adrenal nodule	
☐ no	
☐ indeterminate	
☐ yes	
	Spleen volume: 248.43 cm³
Other findings	
Lymph Nodes	
Retroperitoneal and pararenal lymph nodes	
Other lymph nodes	
Liver	
Billiary tract	
Gallbladder	
Pancreas	
	Normal spleen volume
Spleen	
Stomach and small bowel	

Using clinical-notes skill

Examination		
Exam Type: CT Body Part: Abdomen Contrast: No Laterality: Bilateral		
Clinical Indication		
Unknown		
Comparison		
Prior Studies: No prior imaging available for comparison		
Technique		
Multidetector CT of the abdomen was performed without intravenous contrast. Multiplanar reconstructions were performed.		
Technical quality: Adequate		
Findings		
Spleen		
Consistent with a normal spleen volume (248.43 cm³).		
Additional Regions as Applicable		
- Liver - Gallbladder - Spleen - Pancreas - Kidneys and Adrenals - Gastrointestinal Tract - Peritoneum and Mesentery - Retroperitoneum - Bladder - Pelvic Organs - Vasculature - Lymph Nodes - Bones - Soft Tissues		
Measurements (if applicable)		
Structure	Measurement	Normal Range
Spleen	248.43 cm³	100-315 cm³
Impression		
1. Normal spleen volume (248.43 cm³).		
Communication of Critical Results		
Not applicable		

CHALLENGES

Using Skills

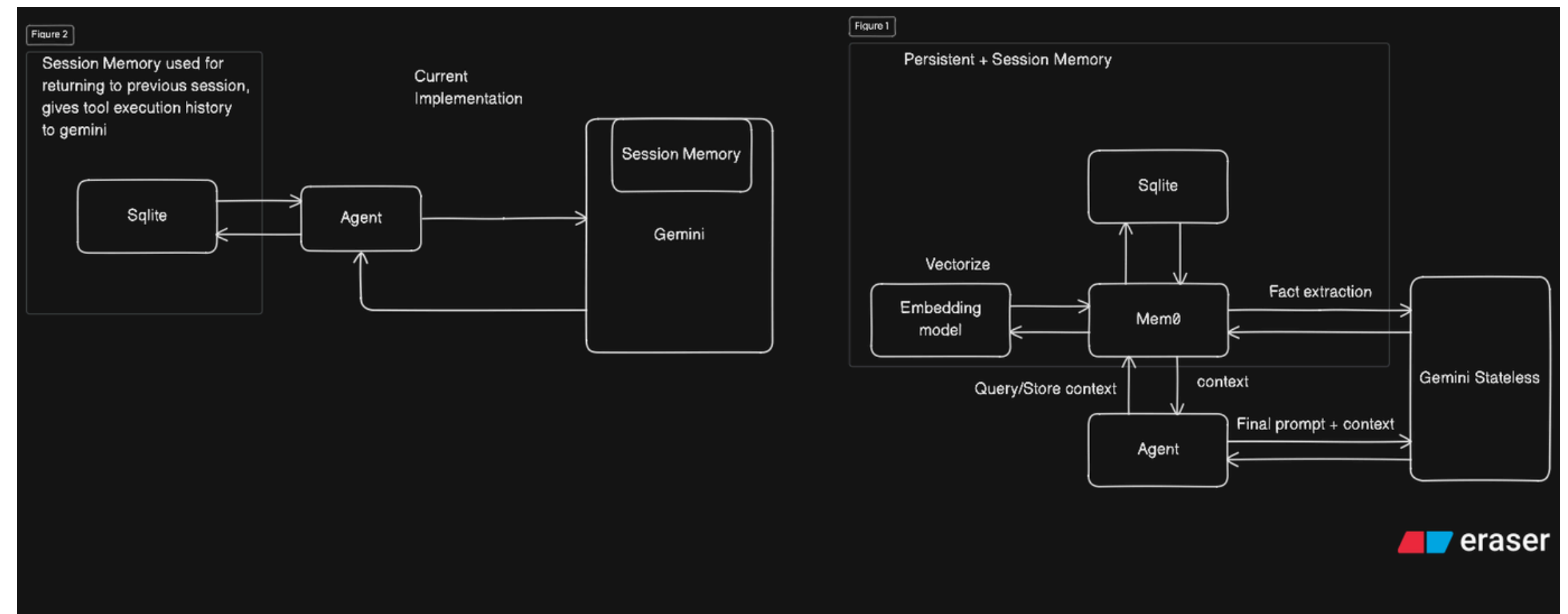
```
],
prompt_token_count=9748,
prompt_tokens_details=[
  ModalityTokenCount(
    modality=<MediaModality.TEXT: 'TEXT'>,
    token_count=9748
  ),
],
total_token_count=9789
) automatic_function_calling_history=None parsed=None
```

After two skill calls:

SKILL.md

references/radiology_report_template.md

```
],
prompt_token_count=21216,
prompt_tokens_details=[
  ModalityTokenCount(
    modality=<MediaModality.TEXT: 'TEXT'>,
    token_count=21216
  ),
],
total_token_count=22014
) automatic_function_calling_history=None parsed=None
```



DEMO

End

